有机化学

概念：

1，价键理论：方向性，饱和性，可以杂化。

2，诱导效应：这种因某一原子或基团的电负性而引起电子云沿着键链向某一方向移动的效应叫诱导效应。电负性强的原子会吸引共价键电子云。造成酸碱性，稳定性等方面的影响。

电负性强的是吸电子基团：X（卤素），-OH，-COOH，

共轭效应： 由于电子离域，使能量降低、键长趋于平均化等现象称做共轭效应

π- π > p – π > σ - π >σ - p....

3，亲电试剂≈路易斯酸，接受电子能力较强的物质：BF3 ，AlCl3，正离子等等...

亲核试剂≈路易斯碱，给出电子能力较强的物质：NH3，RNH2，负离子等等...

4，酸碱性：pKa = -lgKa，描述酸性=对应碱的稳定性=对应碱与H+结合能力

影响因素：对应碱的稳定性（主）

命名

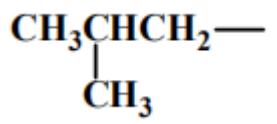
1，官能团优先顺序：先通过官能团优先级确定官能团其他的为取代基，双官能团倒着写在后面

-COOH > -SO3H > -COOR > -COX > -CONH2 > -CN > -CHO > -CO- > -OH > -SH > -NH2 > -C三C- > -C=C- > -OR > -SR > -F > -Cl > -Br > -I > -NO2

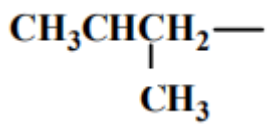
2，较优基团：按分子量排，分子量越大越靠前，第一原子分子量相同比该原子相连的原子，重键按多原子算。-OH>-COOH>-COH>CH_3

3，取代基相连的碳外数1,2,3,4,5对应α，β，γ，δ，ε等

命名



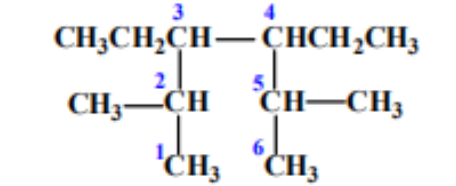
异丁基



仲丁基

1，烷烃（基）：习惯：后数第二个碳上有一个碳的为异某烷（基）

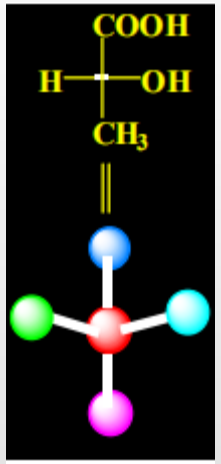
系统：

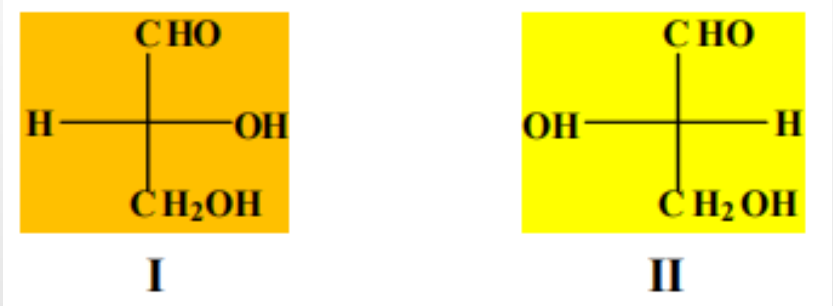


2,5-二甲基-3,4-二乙基已烷

手性：D/L标记，定义甘油醛及其生成物

D/L-某

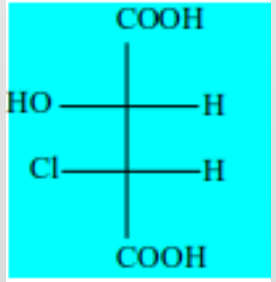




I II

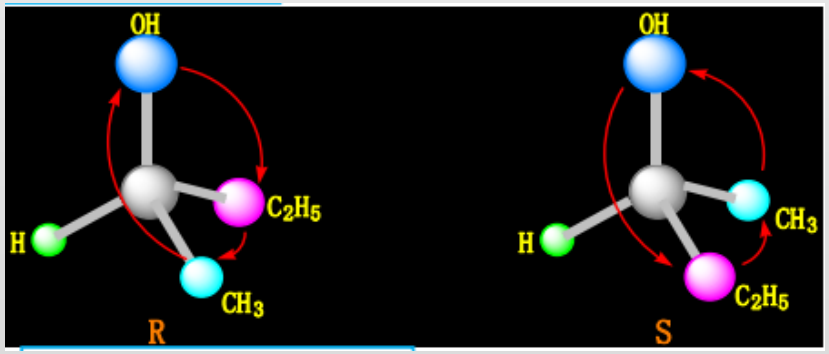
R/S标记，基团优先度顺时针为R

（nR/S，nR/S）-某



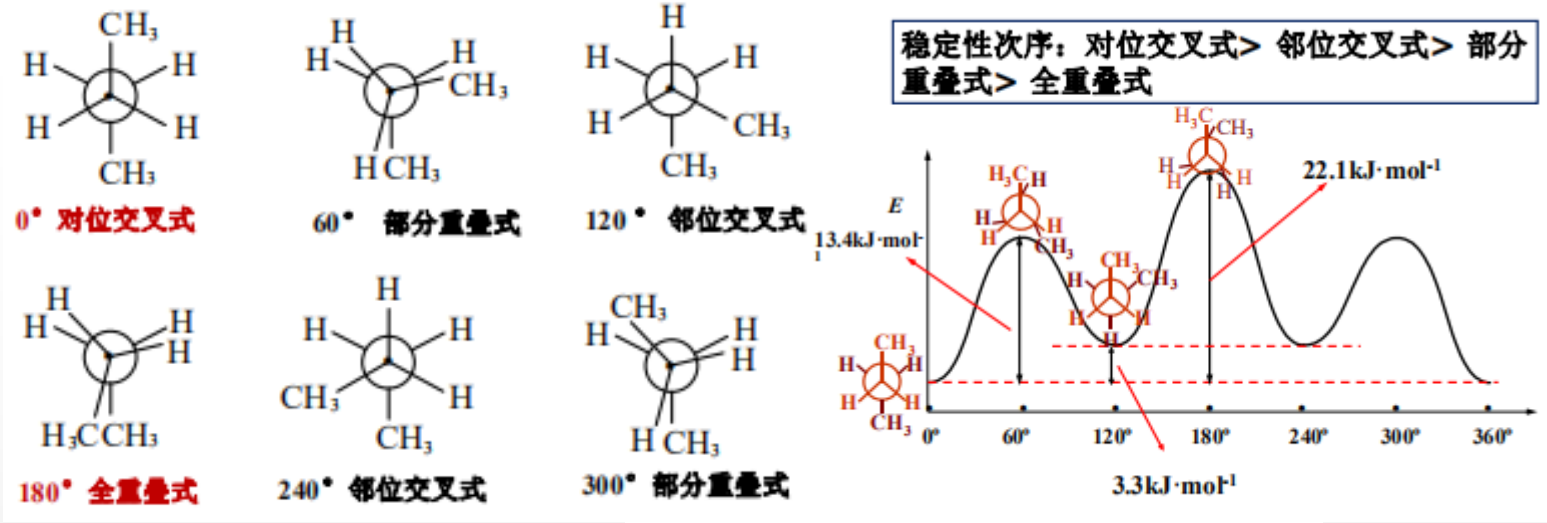
I

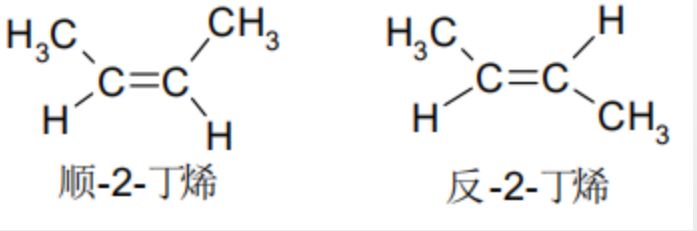
（2R，3R）



命名

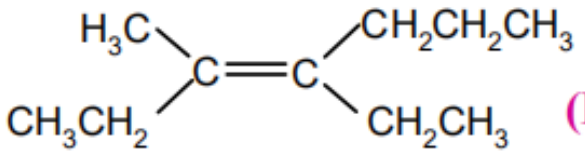
构象异构：



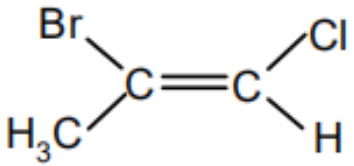


2，烯烃（基）：顺反异构：相同基团在同侧的叫顺式

E-Z标记：较优基团同侧为Z，反侧为E，用（）-标在其他的前面



(E)-3-甲基-4-乙基-3-庚烯



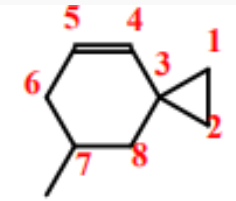
(Z) -1-氯-2-溴丙烯

3，炔烃（基）：同有双键和三键时命为某烯某炔

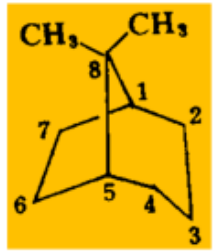
编号要使两者位次数值和最小，若有选择时应使双键的位次最小。

命名

4，脂环烃：双环化合物:螺环化合物：某螺【小环碳原子数，大环碳原子数】某（不算螺碳）其中序号从小环第一个碳出发经过螺碳到大环结束



7-甲基螺[2.5]-4-辛烯



8,8-二甲基双环[3.2.1]辛烷

桥环化合物：某双桥【大环碳原子数，中，小】某（不算桥碳）序号从大环第一个碳出发经过桥碳1到中环经桥碳2到小环回到桥碳1

较优基团同侧为顺

顺-1,4-二甲基环己烷

命名

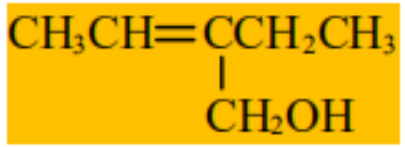
5，芳香族：苯：取代基简单以苯为母体，取代基复杂苯基为取代基。多用临，间，对

苯甲基又称苄基

6，萘：分α和β，同苯为对位

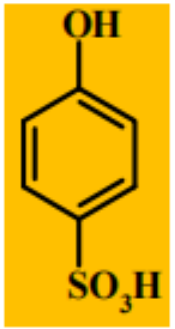
7，卤代烃：习惯命为某卤，系统命名注意取代基次序：R-F-Cl-Br

8，醇：注意双官能团时的次序，



2-乙基-2-丁烯-1-醇

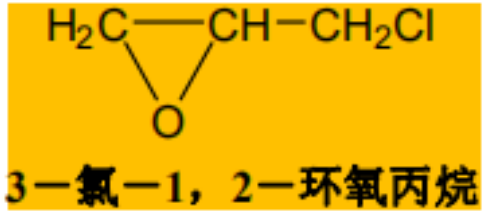
9，酚：注意除卤素，硝基，氨基做官能团外，基本上为取代基



对羟基苯磺酸

对氨基苯酚

10，醚：简单时用取代基小到大为某某醚，复杂时做取代基（-甲氧基），环醚用环氧烷。



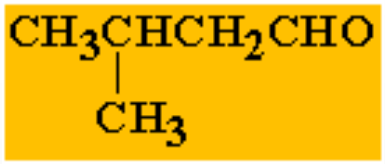
3—氯—1，2—环氧丙烷

$CH_3OCH(CH_3)_2$

2-甲氧基异丙烷

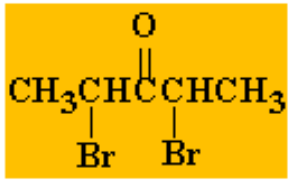
命名

11，醛：同醇

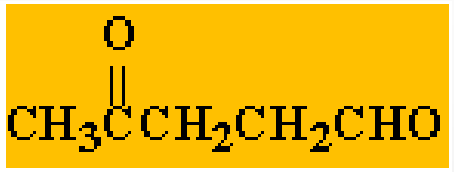


3-甲基丁醛

12，酮：按=O为取代基定编号，做取代基时为“n-氧带”



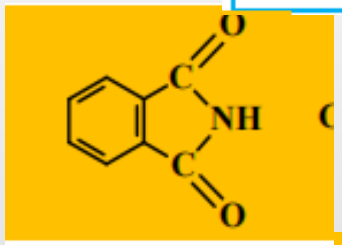
2,4-二溴-3-戊酮
α,α′-二溴-3-戊酮



4-氧代戊醛

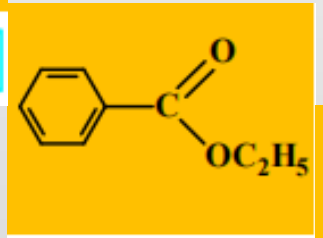
13，羧酸：同醛

14，酰：酸脱-OH为酰，一般为酰卤，酰胺



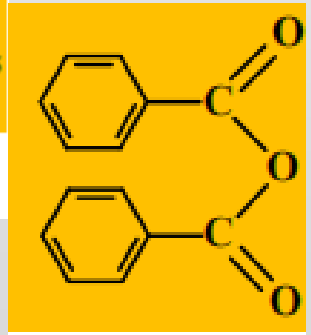
邻苯二甲酰亚胺

15，酯：可以理解为”酰醇”,脱水前的酸和醇去“醇”加“酯”



苯甲酸乙酯

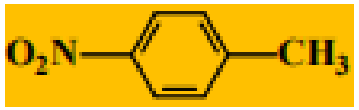
16，酸酐：2mol羧基脱水成1mol酸酐，什么酸脱成什么酸酐



苯甲酸酐

命名

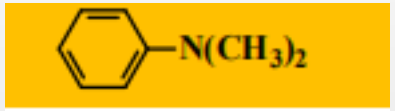
17，硝基化合物：基本上只能是取代基



对硝基甲苯

18，胺：同醇，复杂时为取代基，注意做取代基时应写作“氨基”

当N上连有取代基时，前面标N-



N,N-二甲基苯胺

$CH_3CH_2NH_2$

乙胺

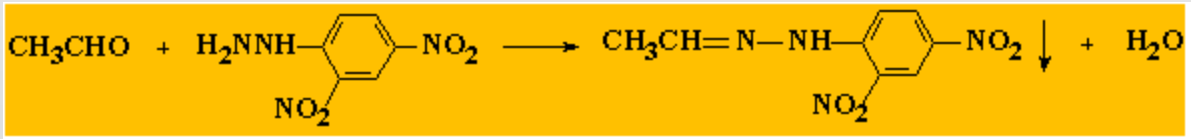
对氨基苯酚

19，铵：当胺成盐时

R_4N^+OH　　$R_4N^+X^-$

季铵碱　　季铵盐

20，肟，腙，脲：由羰基转化



2,4-二硝基苯肼　　乙醛-2,4-二硝基苯腙

黄色沉淀

丙酮肟

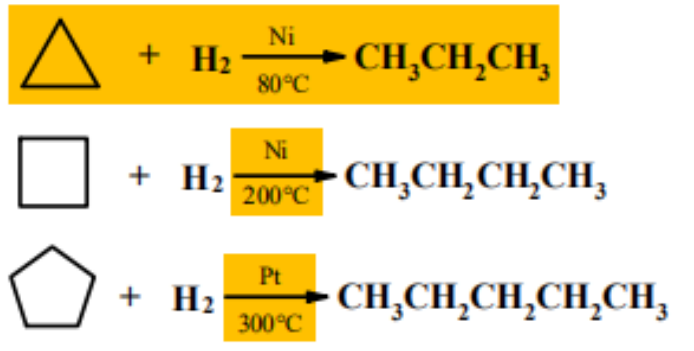
氨基脲　　苯甲醛缩氨脲

合成，引入，转化

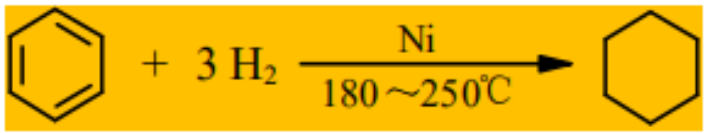
1，烷烃（基）：（1）不饱和烃加氢：Pt，Pd，Ni可完全加成为饱和烃

$$CH_2{=}CH_2 \longrightarrow CH_3CH_3 + H_2$$

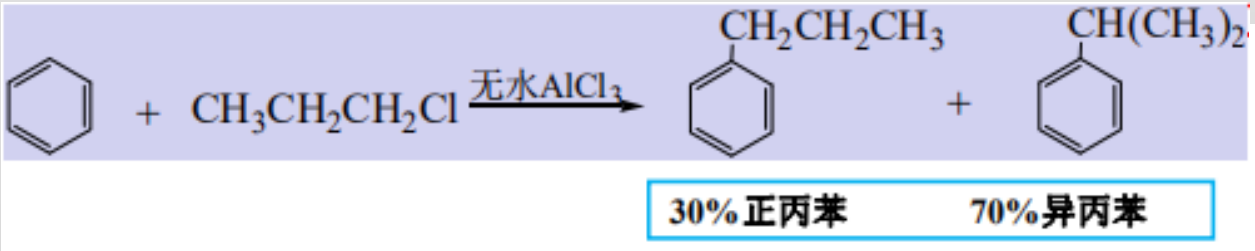
（2）环烷烃开环：Ni，Pt催化，反应条件越来越难

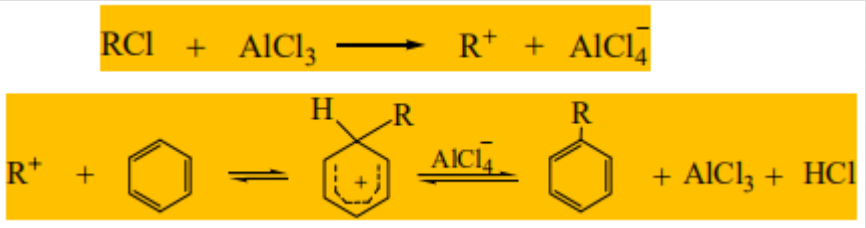


（3）芳香化物加成：



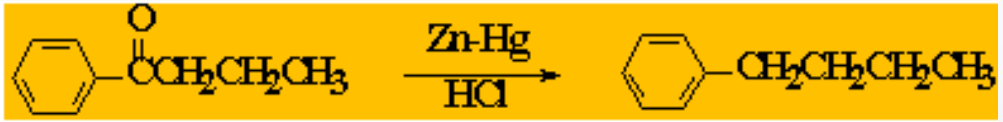
（4）芳香化物烷基化：傅列德尔-克拉夫茨烷基化

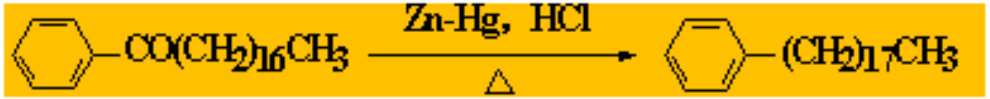




合成，引入，转化

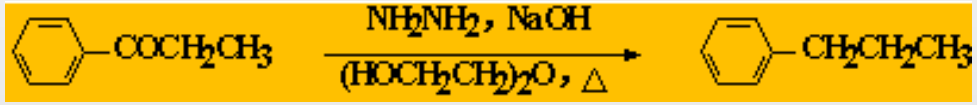
1，烷烃（基）：（5）由酮还原：克莱门森(Clemmensen)还原





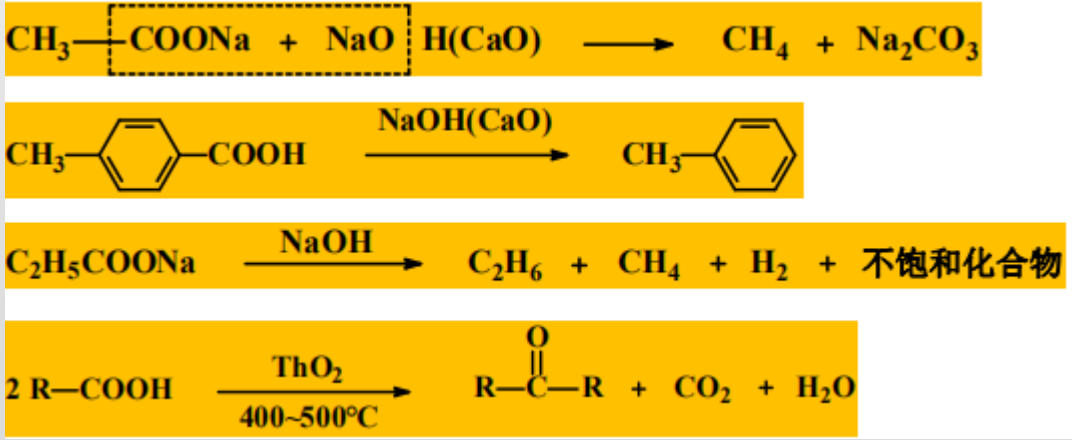
不适合对酸敏感如：含有氨基，同时-NO2也被同时还原。

沃尔夫-凯惜钠-黄鸣龙反应



不适合对碱敏感的如有羧基。

（6）由羧酸脱羧：（一般不用）



合成，引入，转化

1，烷烃（基）：（7）胺的烷基化：

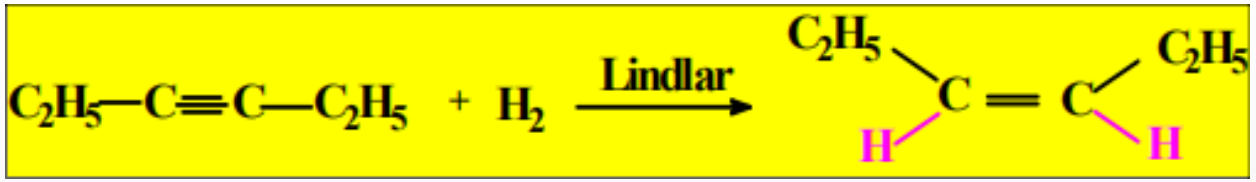
（8）伯卤烷（Br，I）与Na反应：伍兹 –魏蒂希(Würtz- Wittg)反应

无重排

（9）卤代烃与格利雅试剂（-Grignard）：

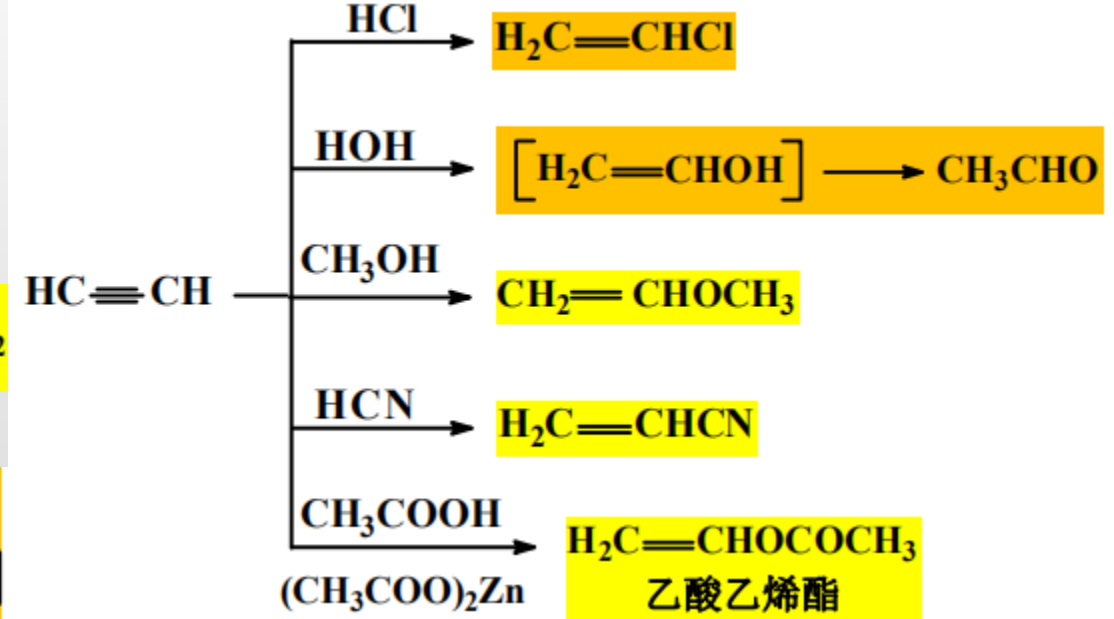
合成，引入，转化

2，烯烃（基）：（1）烷烃裂解，裂化（少）

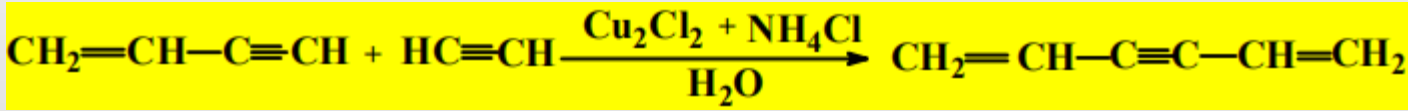


（2）由炔烃加成：与氢可在Lidlar催化剂下可以停留在双键阶段。

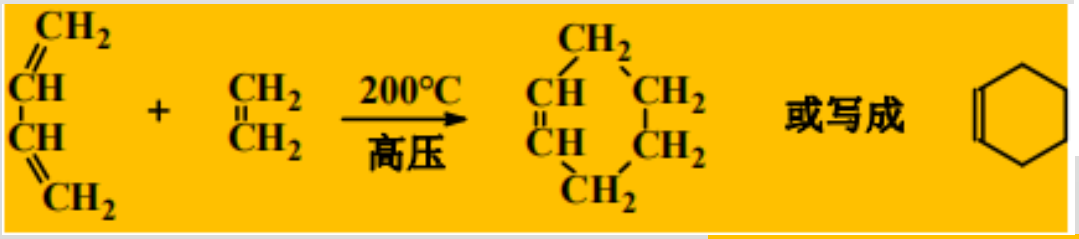
除此还与许多物质加成



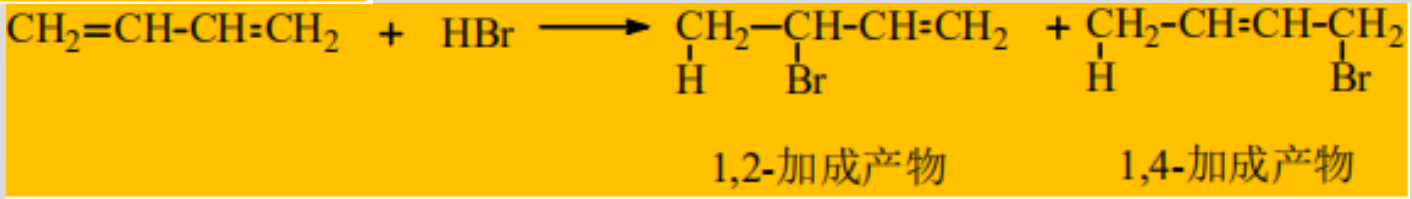
与炔加成（加聚）



狄尔斯-阿尔德(Diels-Alder)反应

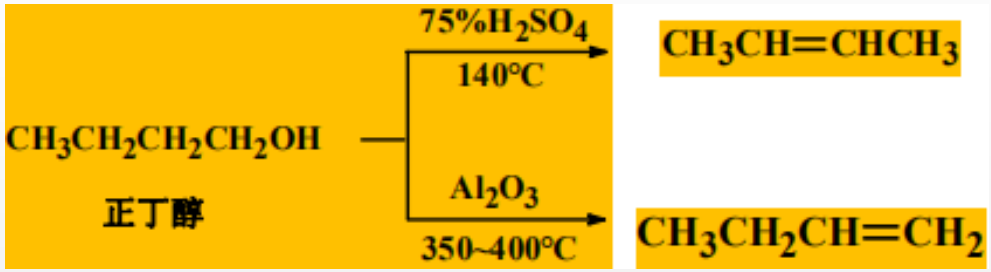


共轭双烯高温加成



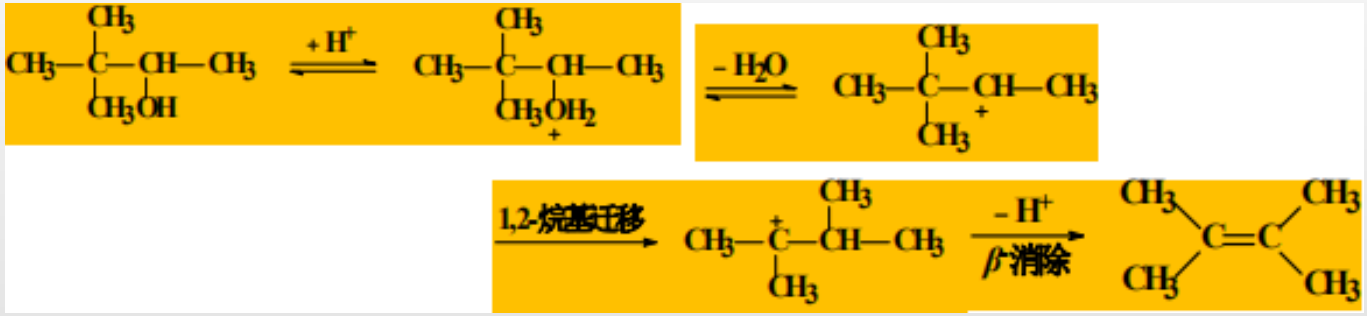
合成，引入，转化

2，烯烃（基）：（3）醇脱水

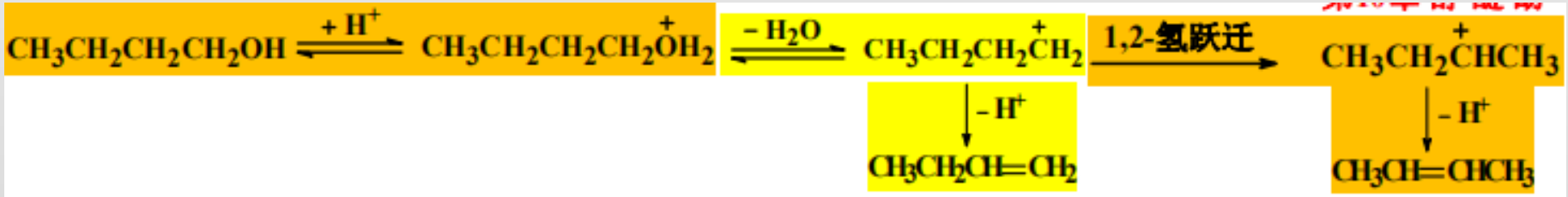


有重排

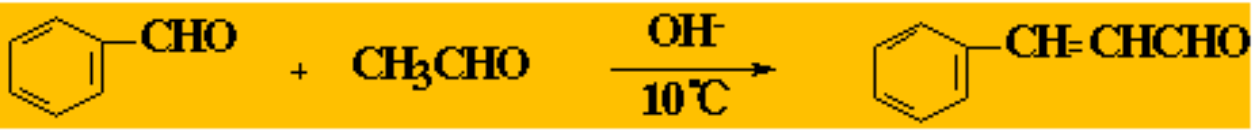
无重排



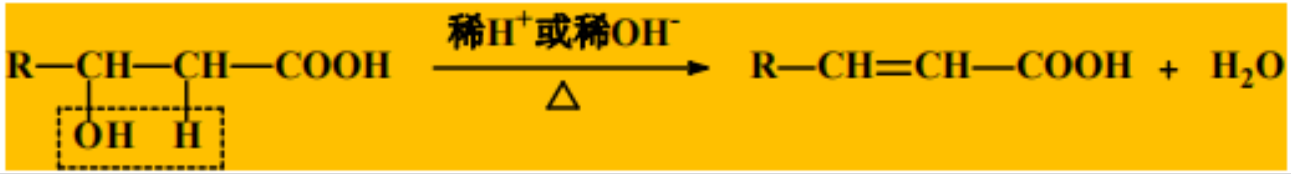
碳正离子稳定性



苯甲醛与有α氢的脂肪酮羟醛缩合



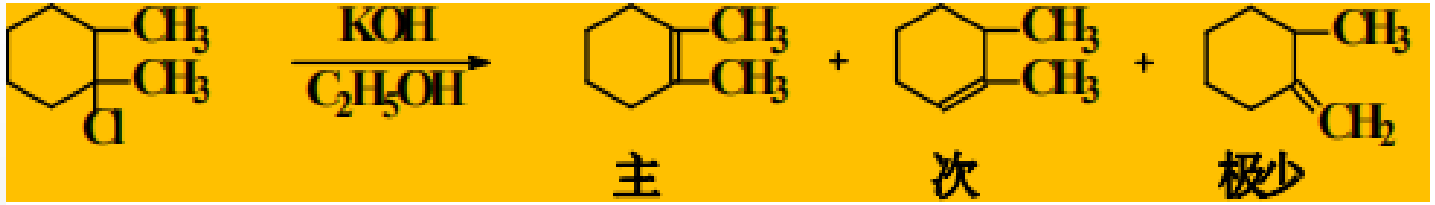
β-羟基酸—分子内脱水生成α，β-不饱和酸

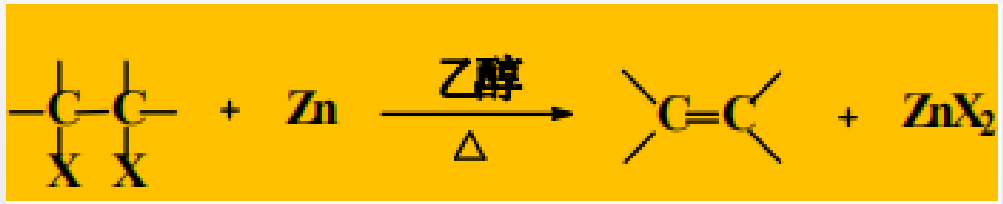


合成，引入，转化

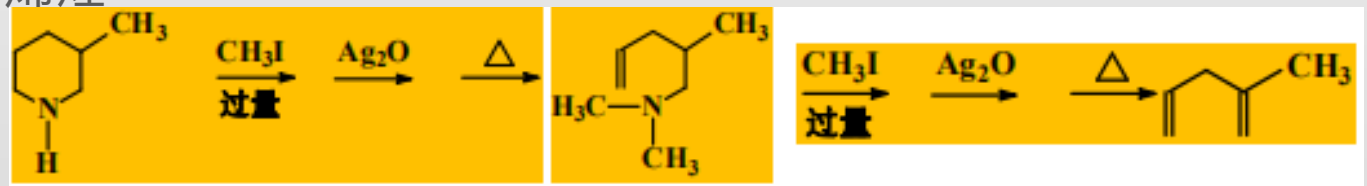
2，烯烃（基）：（4）卤代烃消去：

符合查依采夫(Zaitesev)规则
往氢少的地方消





（5）季铵碱的消除：霍夫曼规则—季铵盐在消除反应中，得到的主要产物为双键上烷基最少的烯烃

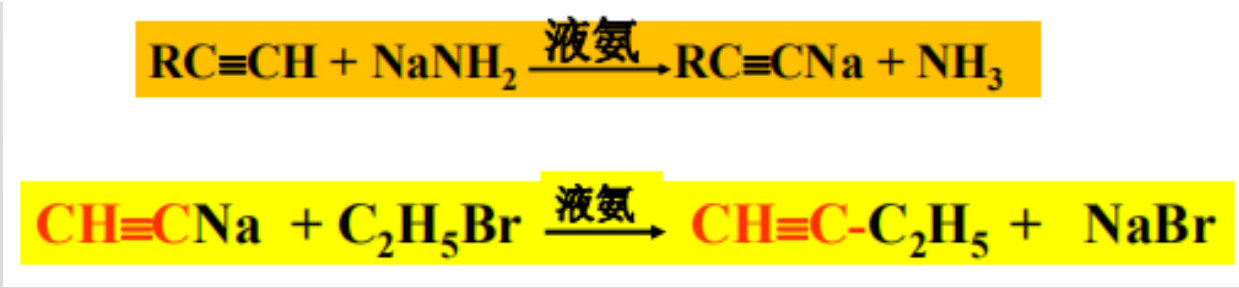


3，炔烃（基）：（1）相邻二卤代烃消去

（2）相邻二醇的消去

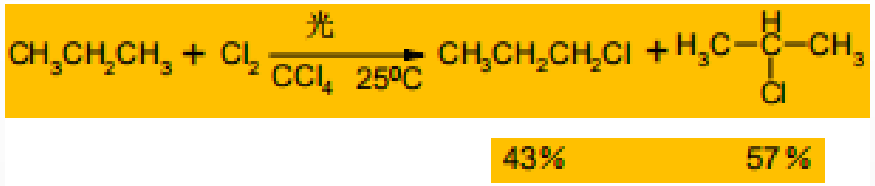
（3）金属炔化物与卤代烃反应：

碳链延长

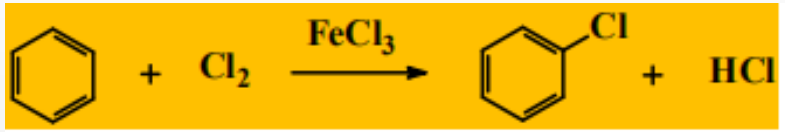


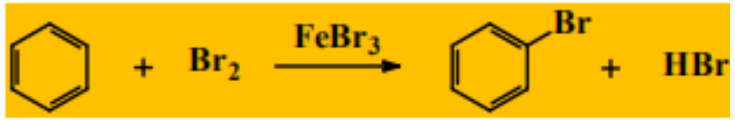
合成，引入，转化

4，卤原子：（1）烷烃的取代：（位置难控，一般不用）

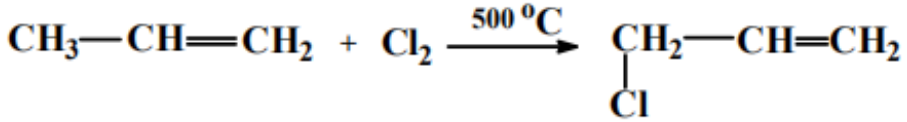


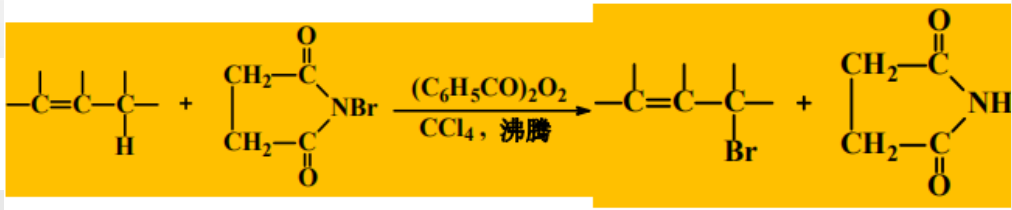
（2）苯环取代：



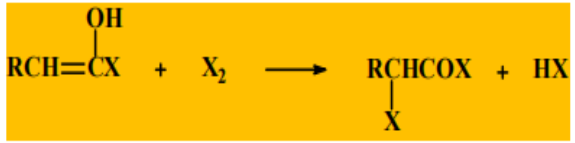


（3）α氢离子活泼性：烯丙基

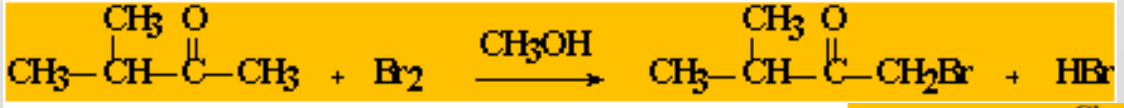




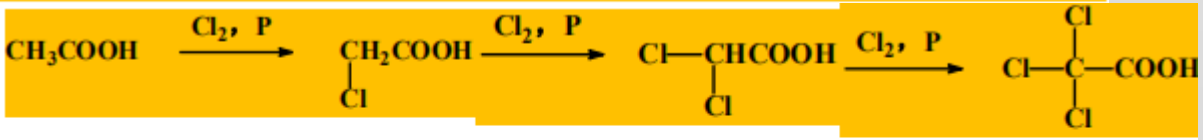
苯甲基光照发生自由基取代



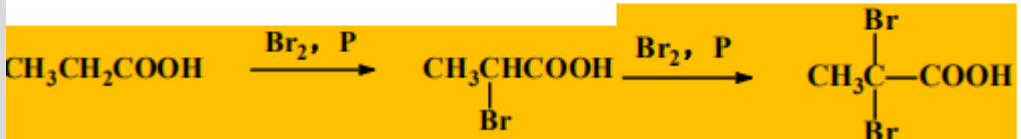
醛酮



羧酸

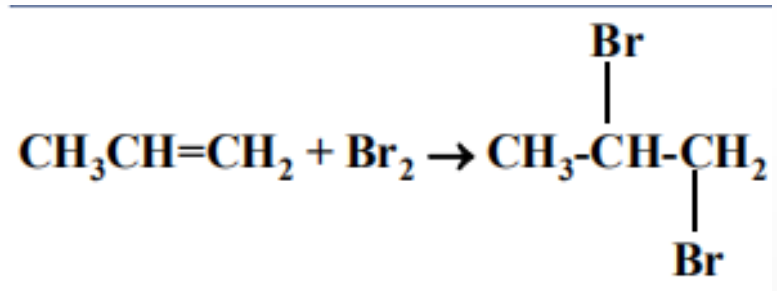


赫尔-乌尔哈-泽林斯基反应

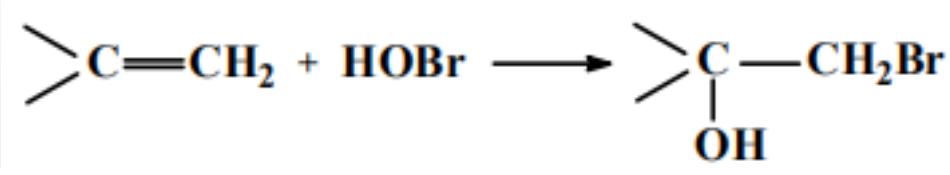


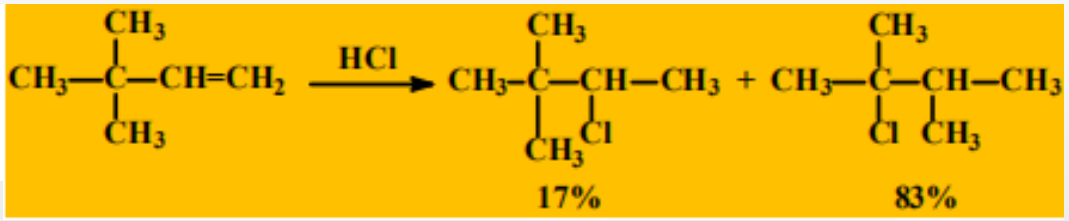
合成，引入，转化

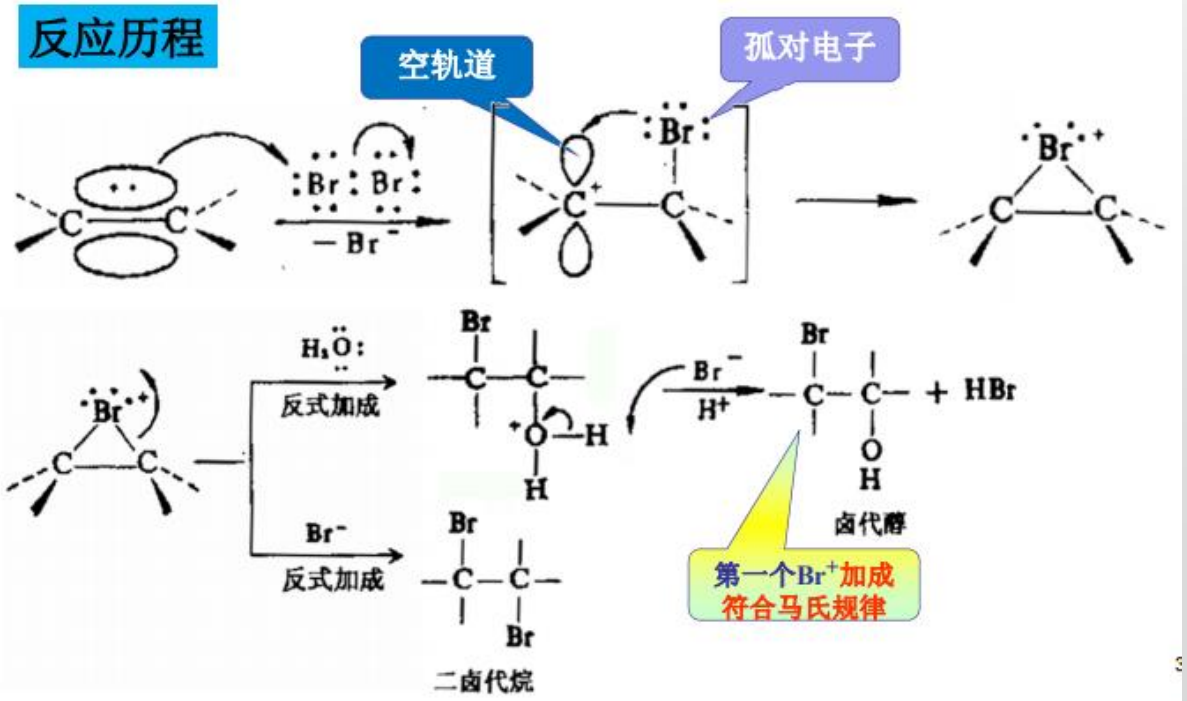
4，卤原子：（4）与不饱和键加成：与卤素发生反式加成

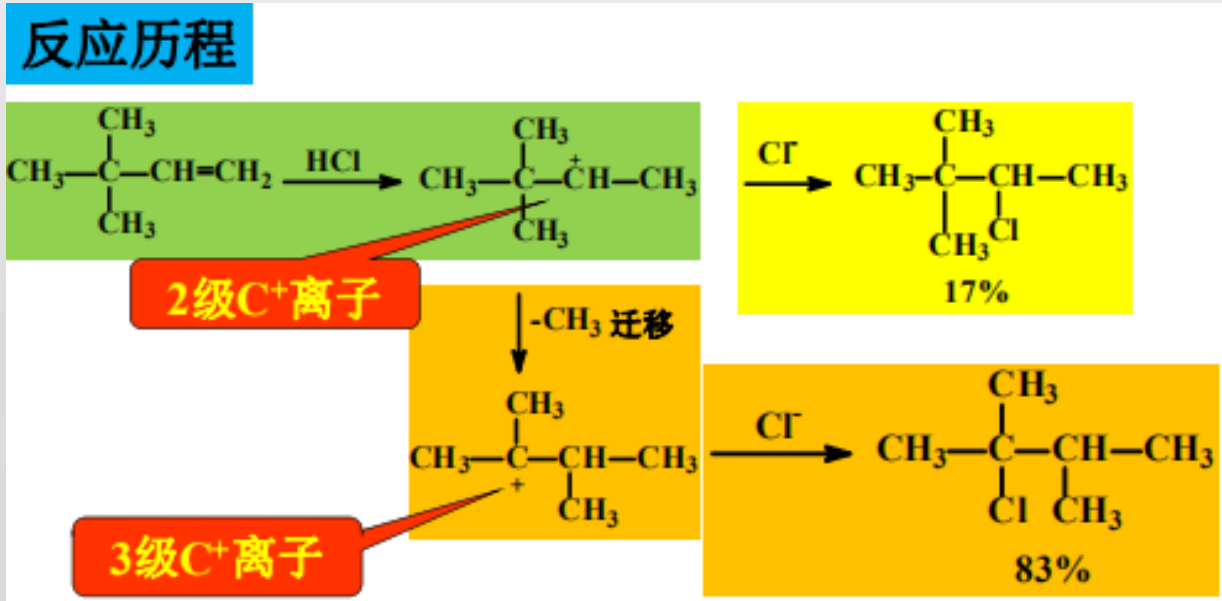


与氢卤酸，次卤酸加成，符合马氏规则（马尔科夫尼科夫Markovnikov）

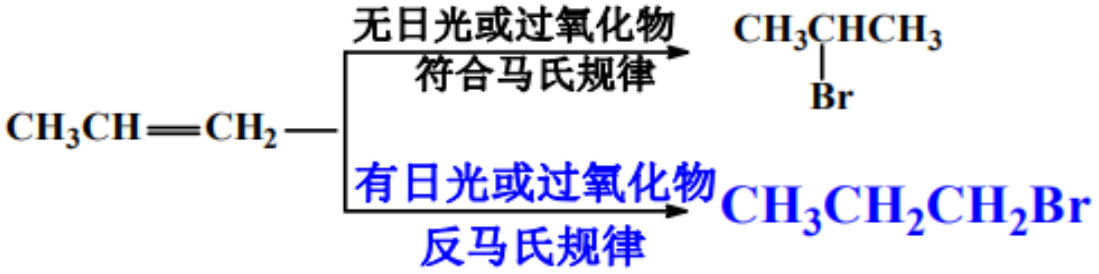








合成，引入，转化



4，卤原子：（4）与不饱和键加成：存在过氧化物，或光照时HBr发生自由基加成为反马氏加成

自由基反应机理

链引发；过氧化物分解为烷氧自由基RO• ，此自由基和HBr作用，引发了自由基溴Br•的生成

RO:OR → 2RO• (烷氧自由基)

RO•+ HBr → ROH + Br• (自由基溴)

链增长：Br•加到烯烃双键上，π键发生均裂，一个电子与溴原子结合成单键，另一个电子留在另一个碳原子上形成另一个烷基自由基。

$$RCH{=}CH_2 + Br\bullet \longrightarrow R\dot{C}HCH_2Br$$

$$R\dot{C}HCH_2Br + HBr \longrightarrow RCH_2CH_2Br + Br\bullet$$

链增长：两个自由基相互结合使链反应终止。

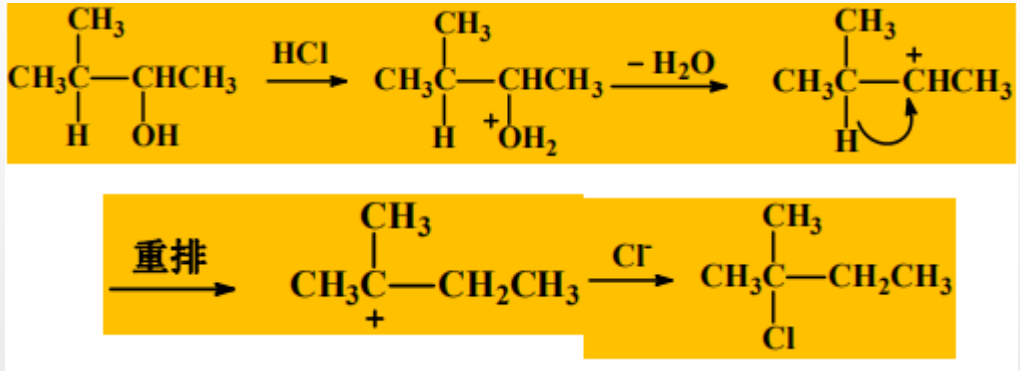
$$Br\bullet + Br\bullet \longrightarrow Br_2$$

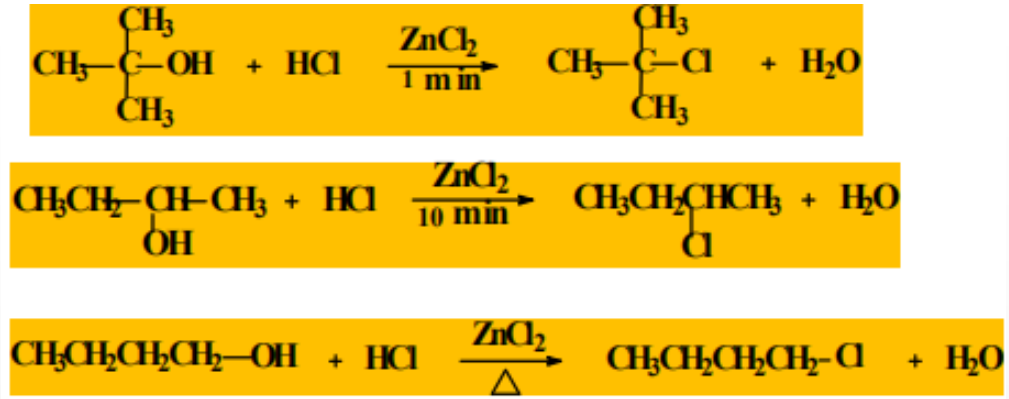
$$R\dot{C}HCH_2Br + R\dot{C}HCH_2Br \longrightarrow \begin{array}{c} RCHCH_2Br \\ | \\ RCHCH_2Br \end{array}$$

$$R\dot{C}HCH_2Br + Br\bullet \longrightarrow \begin{array}{c} RCHCH_2Br \\ | \\ Br \end{array}$$

合成，引入，转化

4，卤原子：（5）由羟基转化：与卤化氢有重排





$$ROH + NaX \xrightarrow{H_2SO_4} RX + NaHSO_3 + H_2O$$

与卤化钠在硫酸催化下反应（伯醇取代，仲醇消去）

与卤化磷反应（Br，I）

$$3\,ROH + PI_3 \longrightarrow 3\,RI + P(OH)_3$$

$(P + I_2$ 或 $Br_2)$

与PCl_5或$SOCl_2$反应（Cl）

$$ROH + PCl_5 \longrightarrow RCl + POCl_3 + HCl$$

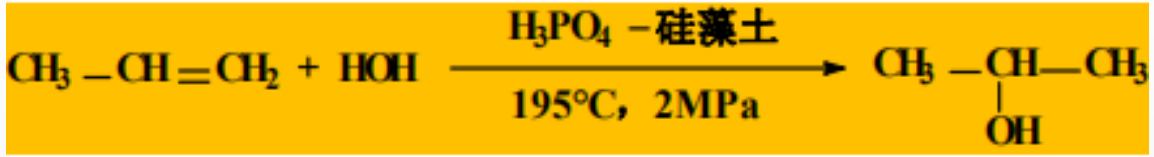
$$ROH + SOCl_2 \xrightarrow{\triangle} RCl + SO_2\uparrow + HCl\uparrow$$

卤素置换

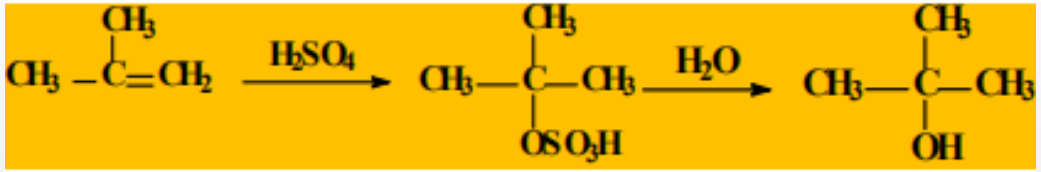
$$RCl + NaI \xrightarrow{丙酮} RI + NaCl$$

合成，引入，转化

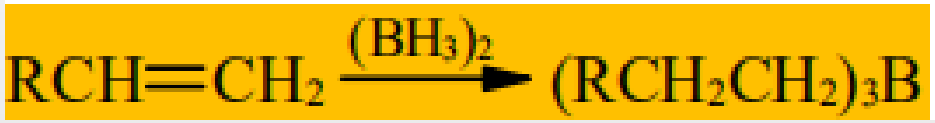
4，羟基：（1）双键加成：



马氏规则

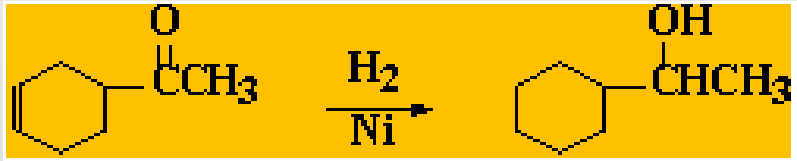


常用，分离烷烃中的烯烃



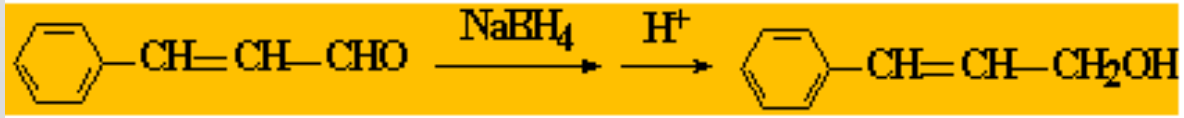
反马氏规则

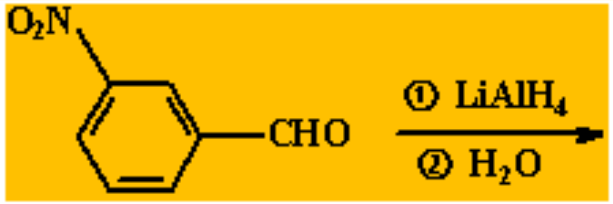
（2）醛酮羧酸还原：

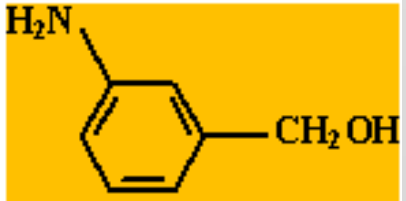


硼氢化钠$NaBH_4$

只还原羰基，不还原不饱和键



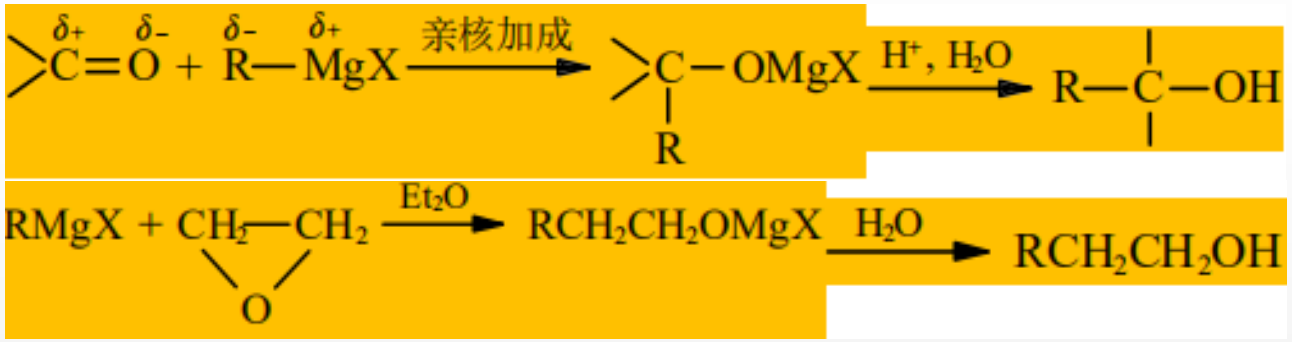




还原羰基，羧基，酯，硝基，氰基
不还原不饱和键

合成，引入，转化

4，羟基：（3）由格利雅试剂制备：

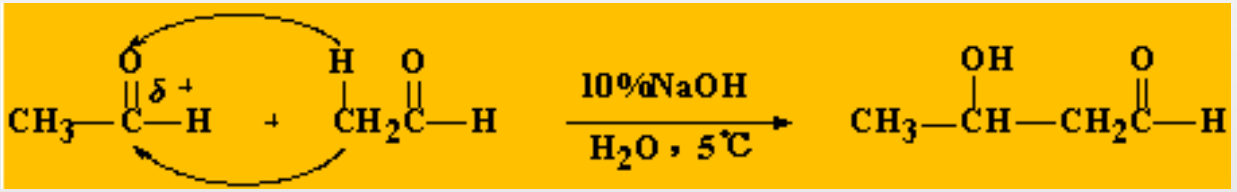


（4）卤素水解：

$$RX + NaOH \rightleftharpoons ROH + NaX$$

（5）羟醛缩合可以得到β-羟基醛：

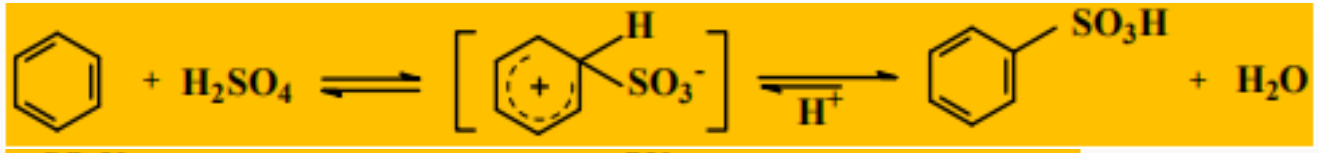
（α碳上有氢时易发生脱水）

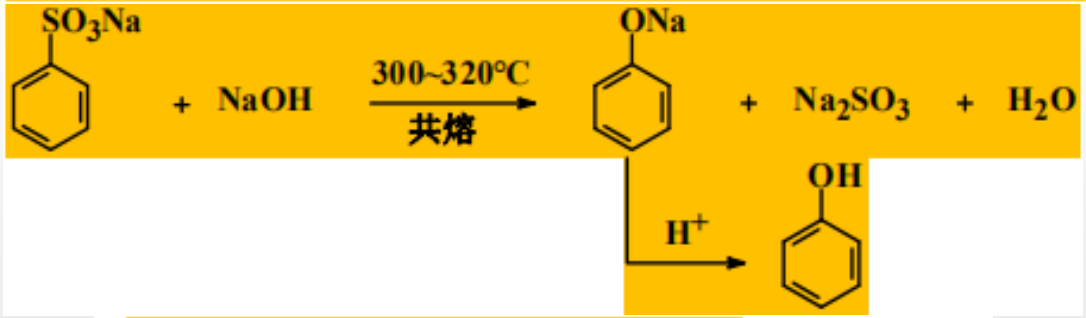


（6）不含α氢的醛易发生歧化：坎尼扎罗（Cannizzaro）反应

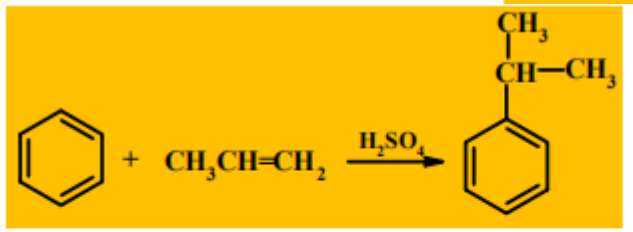
合成，引入，转化

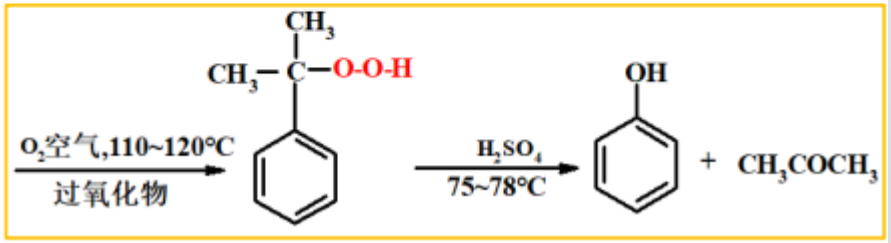
5，酚羟基：（1）磺化碱熔





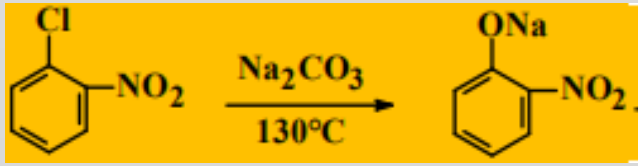
（2） 从异丙苯制备：

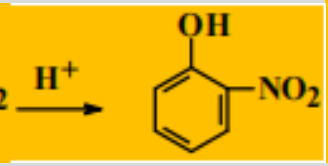


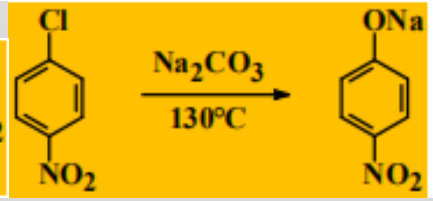


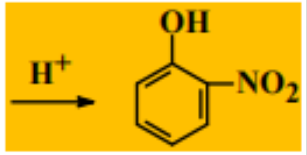
由于形成迈森海默络合物
使反应容易进行

（3）从卤苯制备：









合成，引入，转化

6，醛酮：（1）醇的氧化：

$$CH_3CH_2OH \xrightarrow[260\sim290^\circ C]{Cu} CH_3CHO$$

$$CH_3CH_2OH \xrightarrow{K_2Cr_2O_7+H_2SO_4} CH_3CHO$$

尽快蒸出

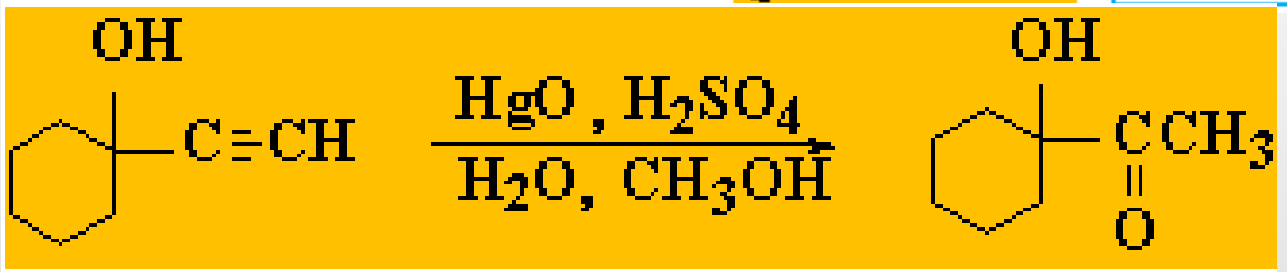
欧芬脑尔氧化法

$$(CH_3)_2C{=}CH(CH_2)_2CH_2OH + CH_3\underset{\underset{O}{\|}}{C}CH_3 \xrightleftharpoons{((CH_3)_2CHO)_3Al} (CH_3)_2C{=}CH(CH_2)_2CHO$$

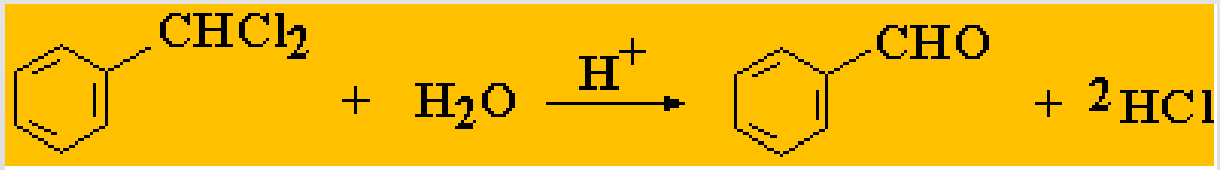
$$\xrightleftharpoons{CrO_3(C_5H_5N)_2}$$

沙瑞特试剂

（2）炔烃水合：

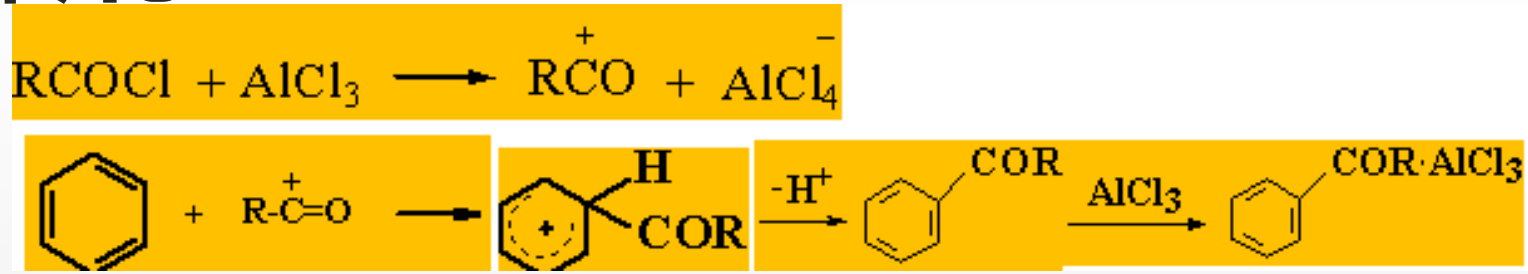


（3）同碳二卤化物水解：

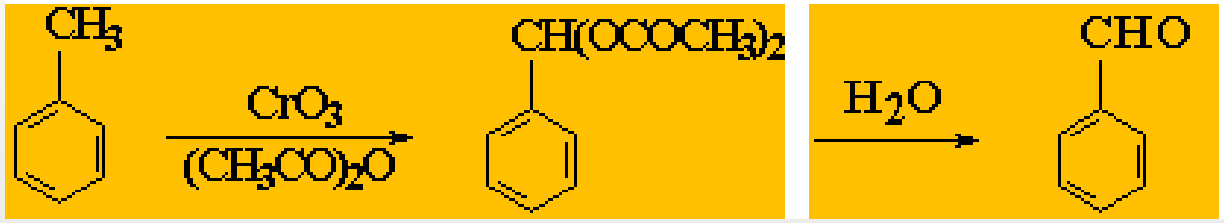


合成，引入，转化

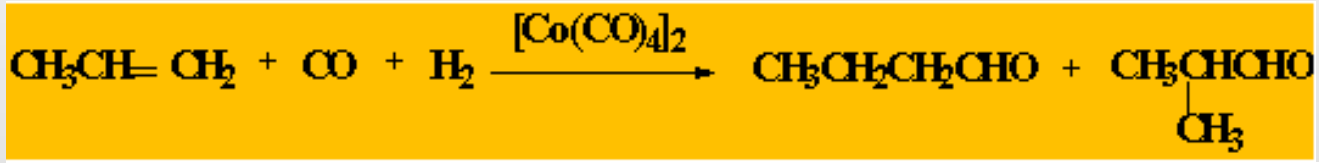
6，醛酮：（4）博-克酰基化



（5）芳烃侧链氧化



（6）羰基合成

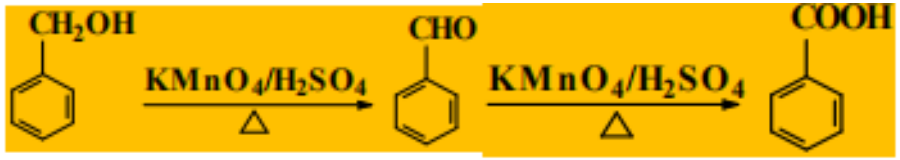


（7）脱羧

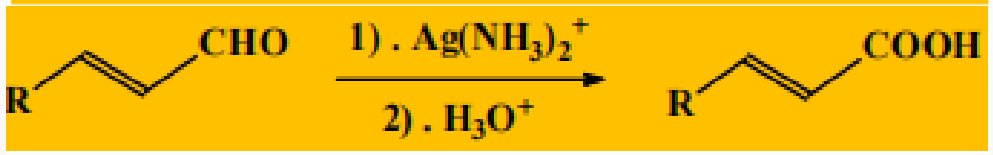
（8）酰卤的还原：罗森门德还原法

合成，引入，转化

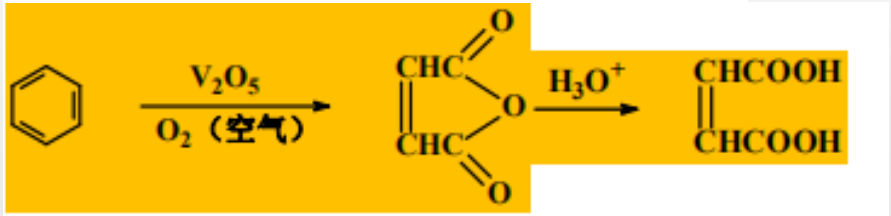
7，羧酸：（1）氧化：伯醇，醛：

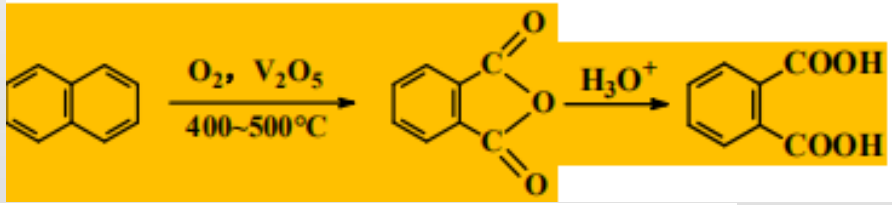


$CH_3(CH_2)_3CHO \xrightarrow[20°C]{KMnO_4/H_2SO_4} CH_3(CH_2)_3COOH$

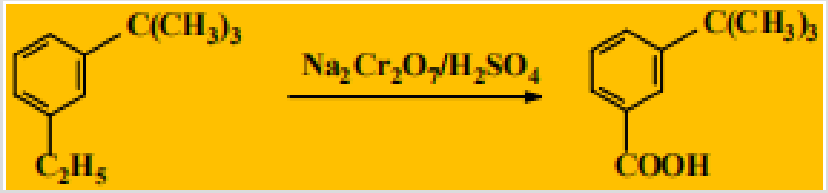


不饱和键氧化：





侧链氧化：



$CH_2{=}CHCH_3 + 3/2\ O_2 \xrightarrow[470°C]{磷钼酸铋} CH_2{=}CHCOOH + H_2O$

合成，引入，转化

7，羧酸：（2）水解：从腈水解：

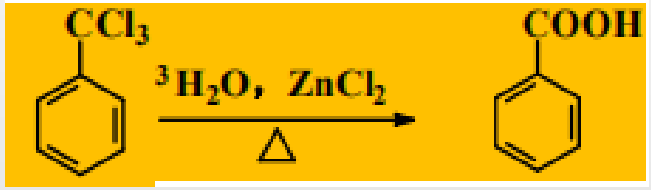
$$CH_2{=}CHCN + H_2O \xrightarrow[\triangle]{H_2SO_4} CH_2{=}CHCOOH + NH_3$$

叔碳不行

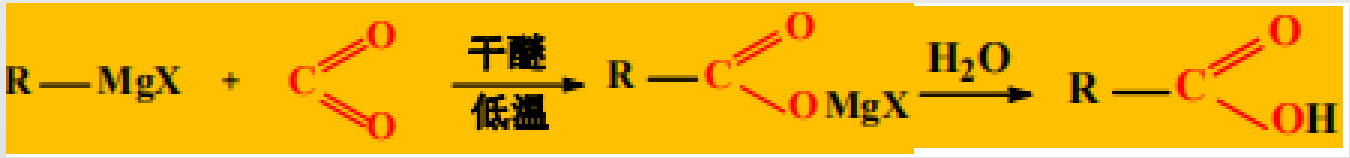
从酯水解：

$$\begin{array}{l} RCOOCH_2 \\ R'COOCH \\ R''COOCH_2 \end{array} + 3H_2O \xrightarrow{H^+} \begin{array}{l} RCOOH \\ R'COOH \\ R''COOH \end{array} + \begin{array}{l} CH_2OH \\ CHOH \\ CH_2OH \end{array}$$

从同碳三卤化物水解



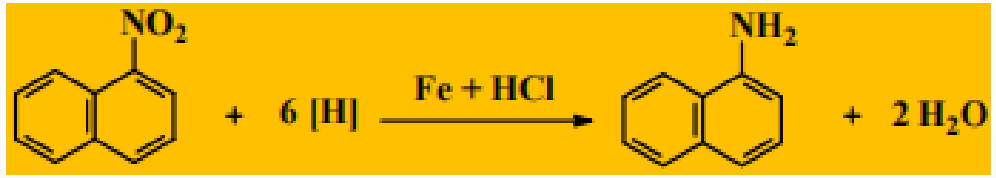
格利雅试剂：



都行

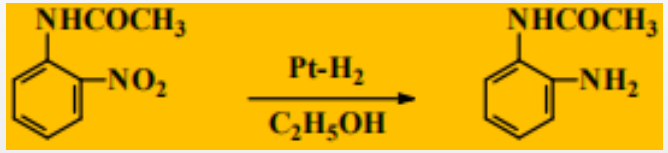
合成，引入，转化

7，胺，氨基，铵盐：（1）硝基的还原：



（2）氨的烷基化：

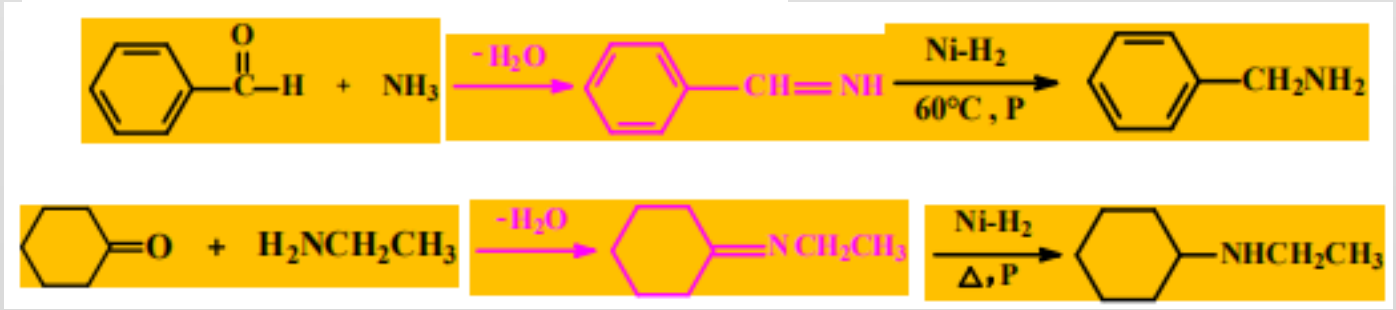
中性



（3）还原：从腈还原：

从酰胺还原：

从醛酮还原：

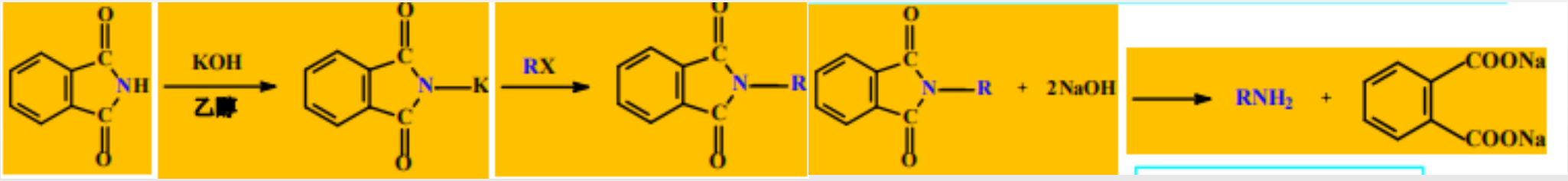


合成，引入，转化

7，胺，氨基，铵盐：（4）霍夫曼胺降级反应：

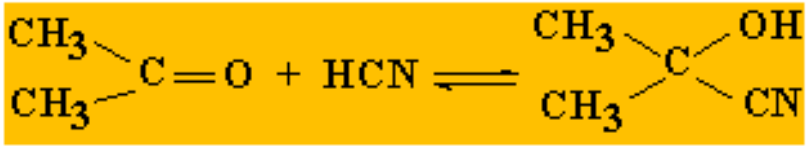
$$RCONH_2 + NaOX + 2NaOH \longrightarrow RNH_2 + Na_2CO_3 + NaX + H_2O$$

（5）盖布瑞尔合成法：



合成，引入，转化

8，腈，氰基：（1）不饱和键加成：碳碳双键，羰基



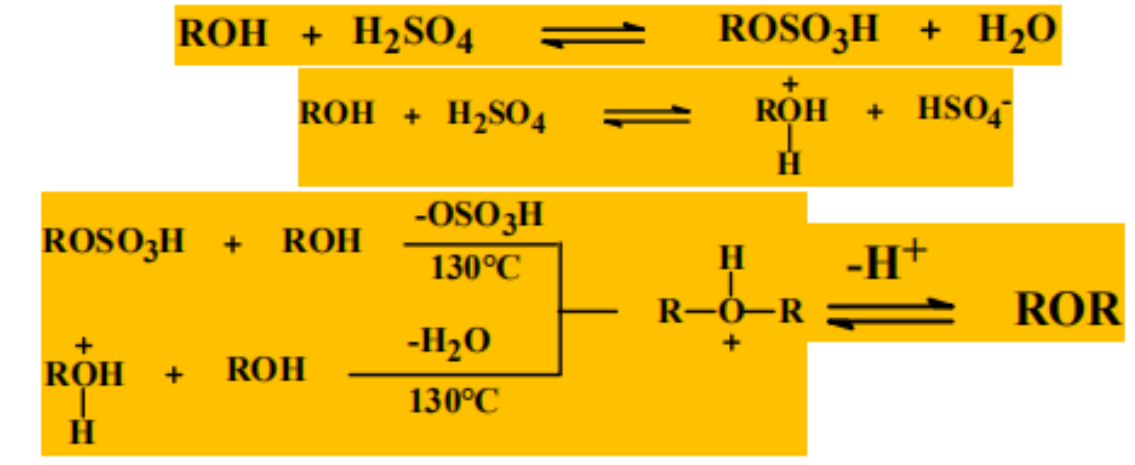
丙酮氰醇

（2）卤代烃与氰化钾加热：

伯卤代烃

$$RX + Na^{+}CN^{-} \longrightarrow RCN + Na^{+}X^{-}$$

9，醚，氧烃基：（1）醇羟基脱水：



除硫酸外，还有芳香族磺酸、氯化锌、氯化铝、氟化硼

$$2\ CH_3CH_2OH \xrightarrow[300^{\circ}C]{Al_2O_3} CH_3CH_2OCH_2CH_3 + H_2O$$

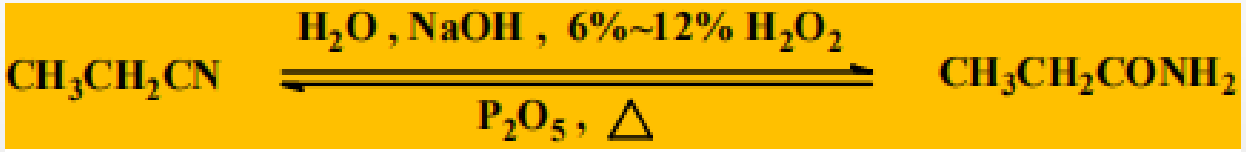
合成，引入，转化

当叔卤烷反应时消去，应用伯卤烷与叔醇钠

9，醚，氧烃基：（2）卤烷与醇钠：

（3）格利雅试剂

10，酰胺：（1）由腈水解：



伯胺，仲胺

（2）由羧酸 生物制备：

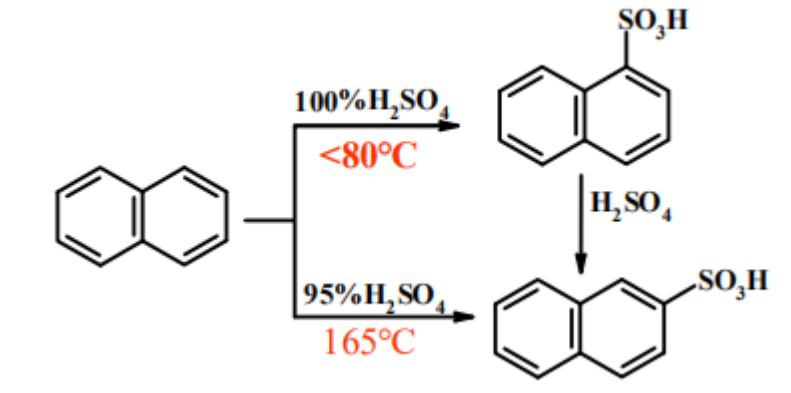
11，酯：（1）由羧酸，酸酐和羟基：

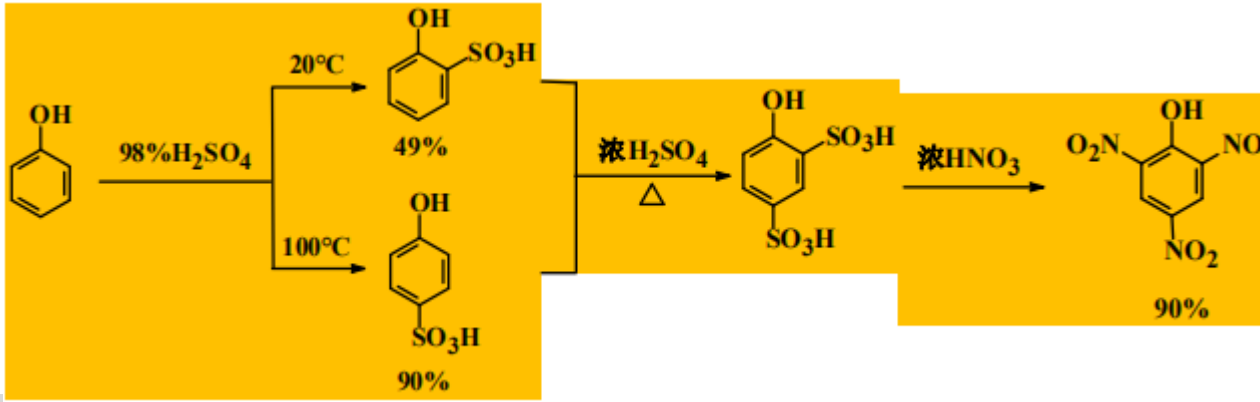
（2）由酰卤与羟基：

占位

12，磺酸：（1）

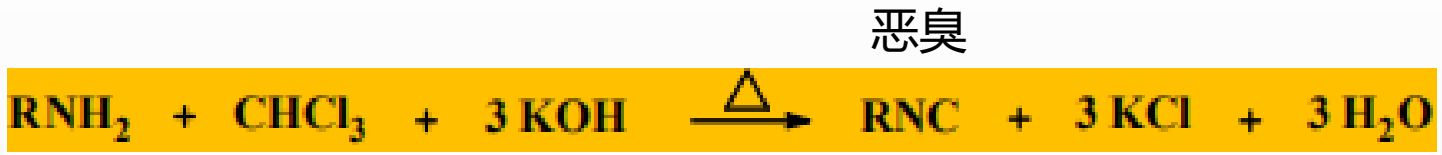
由苯取代





鉴别

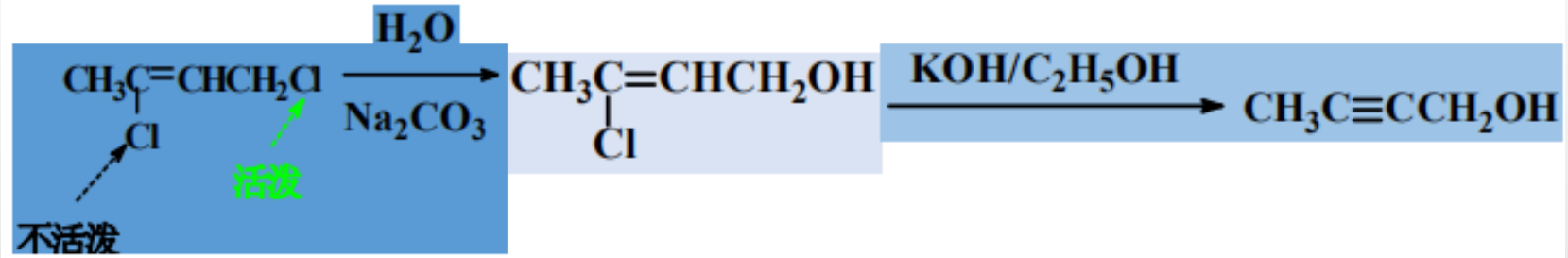
伯胺：异腈反应：



卤原子：

颜色

丙烯中的卤原子：

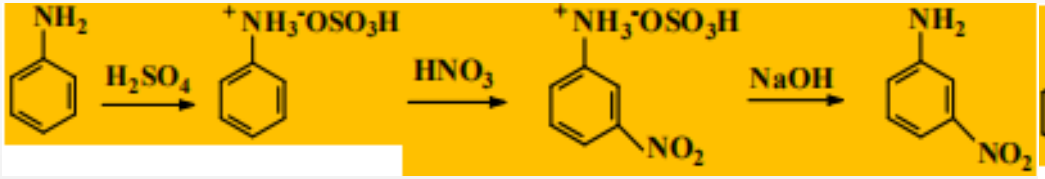


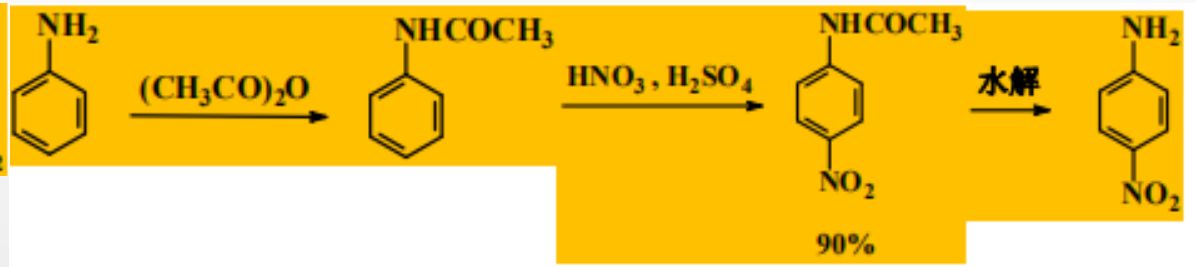
醇，醚，酚：

醇，醛，酸：

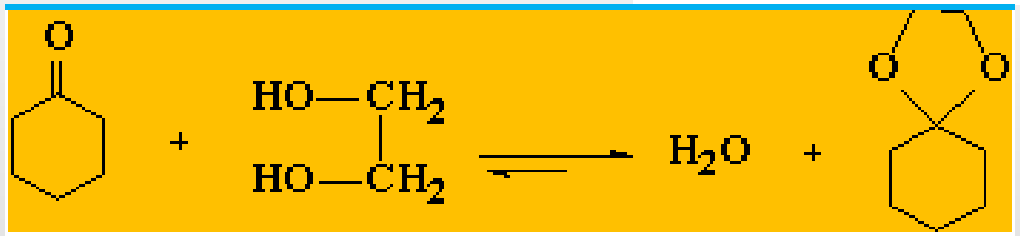
保护

伯胺：酰基化-水解，磺化-水解





醛基：缩醛-水解



酚羟基：酰基化-水解

硝基：

酸碱性

1，酸性排序：XCH2COOH > CH3COOH > RCH2COOH > CH3OH　　诱导效应吸电子基时负离子稳定

/\　　/\

临/对硝基苯甲酸>间硝基苯甲酸>苯甲酸>临/对硝基苯酚>间硝基苯酚>苯酚

邻对位为吸电子诱导效应+共轭效应使对应的碱更稳定，间位只有吸电子的诱导效应。

2，碱性排序：

$(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N >$ 苄胺 $> NH_3 >$ 苯胺

取代基为供电子基碱性变强，吸电子基碱性变弱

电子效应看烷基多碱性强，溶剂效应看烷基多碱性弱

苯上有吸电子基碱性变弱，供电子基碱性变强